

Contact

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(LinkedIn)

portfolio-preethia.vercel.app
(Portfolio)

Top Skills

GraphQL

Agentic Workflows

Retrieval-Augmented Generation
(RAG)

Certifications

Academy Accreditation - Generative
AI Fundamentals

Introduction to DevOps

Continuous Integration and
Continuous Delivery (CI/CD)

Preethi A

M.S. Computer & Information Sciences @ SDSU | Software
Engineer (.NET, SQL, Backend) | Data Analytics (Power BI, ETL,
Excel, Python) | ML/AI Research | IBM DevOps Certified
Manchester, Connecticut, United States

Summary

Hi, I'm a recent M.S. graduate in Computer and Information Sciences from South Dakota State University, where I worked as a Graduate Research Assistant on making video AI systems faster, more efficient, and easier to deploy in real world settings, with a focus on video anomaly detection for real time video understanding. My thesis work centered on RegiGrow, a continual learning system for video anomaly detection that I designed and deployed. It uses the ImageBind encoder with MoELoRA adapters so the model can adapt to new video domains over time without forgetting what it already learned. I built an entropy based expert growth mechanism that detects visual domain shifts on demand, trained and evaluated it across 45,000+ video frames on SLURM HPC, and put together a full evaluation framework measuring Backward Transfer, Forward Transfer, and Forgetting across multiple benchmarks. The goal throughout was to keep things lightweight, practical, and reproducible. Alongside my master's studies, I worked as a Software Engineer in the Enterprise Applications division at South Dakota State University. As part of the development team, I contributed to campus wide applications, including a project called the USDA Capacity Reporting system. My day to day work spanned both frontend and backend development, database design and SQL, debugging and testing, and writing clean, production ready code. I collaborated closely with other developers using GitLab for version control, followed an agile workflow with a ticketing system, and helped deliver accessible, scalable solutions used across the university. I enjoy working at the intersection of AI research and real software systems, and I'm currently exploring opportunities in software engineering, data analysis, and research analyst roles where I can contribute, keep learning, and help build meaningful, real world solutions. Let's connect.

Experience

South Dakota State University

1 year 4 months

Graduate Research Assistant

September 2025 - May 2026 (9 months)

Conducting research on fast and sustainable video anomaly detection using continual learning techniques, incorporating LoRA and Mixture of Experts (MoE) approaches to enhance model efficiency and adaptability. Assisting in experiments, data analysis, and development of innovative deep learning methods for real time video understanding.

Graduate Teaching Assistant

August 2025 - August 2025 (1 month)

United States

I am conducting thesis research on “Fast and Sustainable VAD with Continual Learning,” focusing on real-time Video Anomaly Detection (VAD). Alongside my research, I contribute as a Graduate Teaching Assistant, supporting courses, mentoring students, and assisting in academic development within the Computer & Information Sciences department.

Student IT Programmer

February 2025 - August 2025 (7 months)

Brookings, South Dakota, United States

- As a full stack developer on campus wide enterprise applications, including the USDA Capacity Reporting project, as part of the Enterprise Applications dev team at South Dakota State University.

- Built and maintained both frontend and backend features using C#, .NET, JavaScript, HTML, CSS, Bootstrap, and SQL to support scalable and accessible reporting systems used across the university.

- Designed and optimized complex SQL queries and CTEs to support reporting, data processing, and overall system performance.

- Developed responsive and accessible user interfaces with HTML, CSS, JavaScript, ASP.NET, and Bootstrap, and created dynamic online forms using Formsite while ensuring accessibility compliance.

- Worked closely with other developers using GitLab and Git Bash for version control, followed an agile, ticket based workflow, and regularly contributed to

debugging, testing, documentation, and delivering clean, production ready code.

Education

South Dakota State University

Master of Science - MS, Computer and Information Sciences and Support Services

Annamacharya Institute of Technology & Sciences, Tirupati

Bachelor of Technology - BTech, Data Science and Artificial Intelligence · (November 2020 - May 2024)